Factorized Representation Learning for Data-Scarce Prediction: Application to Monsoon Onset

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Abstract

Forecasting the onset of the Indian summer monsoon over Kerala (a state in south India) weeks in advance is crucial for water resource management and agricultural management in India. The associated machine learning problem using reanalysis data as input presents a fundamental challenge: inputs are high-dimensional and abundant, but supervision is extremely scarce, with only one labeled sample per year. This makes standard end-to-end deep learning statistically ill-posed and prone to overfitting. We address this challenge with a factorized learning framework that decouples spatial representation learning from temporal forecasting. The key idea is to exploit the temporal structure to transform N labeled samples into $O(NT)$ effective training instances by using self-supervised learning on temporally decomposed inputs. We learn time-invariant spatial representations from pre-monsoon climate fields using a self-supervised objective based on Bootstrap Your Own Latent, and train a lightweight temporal model that iteratively refines onset predictions through a novel delta-based formulation. This factorization acts as an implicit regularizer, enabling stable learning under extreme data scarcity while preserving nonlinear spatio-temporal modeling capacity. Using ERA5 reanalysis data of eight crucial atmospheric and ocean variables in the pre-monsoon period, the proposed method gives a weekly forecast of the Monsoon onset date starting from the first week of February (approximately 4 months of lead time). The forecast issued on 1 May achieves a root mean squared error of 1.63 days in a fixed train–test split and 1.83 days in a leave-one-year-out evaluation, reducing the error by more than 50% relative to an operational statistical baseline based on principal component regression. Overall, the proposed approach converts an ill-posed forecasting problem into a tractable one, enabling accurate prediction in extreme low-data regimes.

1 Introduction

The Indian Summer Monsoon (ISM) is one of the most influential climate systems on Earth, yet predicting its onset remains fundamentally challenging. Dynamically, the Monsoon onset over Kerala (MoK) is a singular event that occurs with a mean of 1 June and a standard deviation of 8 days. This event marks the beginning of the Monsoon season. The machine learning task is to predict the MoK date using a sequence of multi variable weather data as early as possible (with 30 to 120 days lead time). Typically, the pre-monsoon season is from the month of February. Thus, our goal is to predict the onset using data from 1 February, at weekly intervals, all the way to the last week of April.

For machine learning, the challenge is that high quality reanalysis data and onset data are available from 1951 onward, leading to only 75 labeled samples for model development. In this regime,

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models must learn complex, nonlinear interactions across the ocean–atmosphere system from sparse supervision. High-capacity deep learning models tend to overfit, while statistical approaches such as principal component regression (PCR) lack the expressivity to capture these interactions. The Indian Meteorological Department uses a PCR based method to forecast the onset by 01-15 May and achieves a RMSE of 4 days with a lead time of approximately 15 days [45]. Numerical weather prediction systems offer strong performance but require expensive data assimilation and simulation pipelines. Consequently, a fundamental disconnect persists between rich modeling expressiveness and the capacity to learn robustly in the face of extreme data scarcity. The core question we address is: how can we train dependable predictive models with high dimensional inputs but extremely limited labeled instances?

In this work, we address this challenge through a factorized learning framework that decouples spatial representation learning from temporal prediction. The key idea is to exploit abundant unlabeled spatio-temporal reanalysis fields to learn invariant spatial representations, while reserving limited supervision for target prediction. Concretely, we learn spatial embeddings using self-supervised learning on individual temporal snapshots, effectively increasing the number of training samples via temporal decomposition, and model the progression toward monsoon onset using a lightweight temporal model over these representations.

We reformulate the task as an iterative refinement process. Instead of predicting the onset date, the model produces incremental updates over time, progressively refining its estimates as additional observations become available. This formulation aligns with the gradual evolution of large-scale climate processes and enables meaningful predictions even from partial input sequences.

We evaluate the proposed framework on ERA5 reanalysis data for predicting monsoon onset over Kerala at a lead time of approximately 30 days. Under both fixed train–test and leave-one-year-out (LOYO) evaluation protocols, our method substantially outperforms operational statistical baselines, reducing prediction error by more than 50%. The model also demonstrates strong temporal generalization and retains predictive skill under truncated inputs, indicating effective utilization of early signals.

Our results show that decoupling representation learning from temporal modeling provides a principled and effective approach for prediction in data-scarce settings. Beyond monsoon onset, this framework offers a general strategy for learning from high-dimensional spatio-temporal data when supervision is limited.

Research Questions. We investigate the following questions:

- Can self-supervised representation learning overcome extreme data scarcity in climate prediction tasks with fewer than 100 labeled samples?
- Does decoupling spatial representation learning from temporal modeling improve generalization across unseen years?
- Can iterative refinement enable stable and accurate early predictions from partial observation sequences?

Contributions. Our main contributions are:

- We propose a factorized learning framework that separates spatial representation learning from temporal prediction, enabling robust learning under extreme data scarcity.
- We introduce a self-supervised pretraining strategy that leverages temporal decomposition to significantly increase the effective number of training samples.
- We develop an iterative refinement-based temporal model that produces stable, progressively improving predictions and supports flexible lead-time forecasting.
- We demonstrate substantial improvements over operational baselines on monsoon onset prediction, along with strong temporal generalization under strict evaluation protocols.

2 Background and Related Work

2.1 Monsoon Dynamics and Onset Mechanisms

The Indian Summer Monsoon (ISM) is a large-scale coupled ocean–atmosphere system that governs the hydrological cycle over the Indian subcontinent, contributing nearly 70–90% of annual rainfall and exerting a strong influence on agriculture, water resources, and economic activity. Its onset over Kerala (MOK) marks a rapid transition from a dry pre-monsoon regime to a convectively active state and typically occurs around early June with relatively low interannual variability. Despite this apparent regularity, the onset is not governed by a single process but emerges from the interaction of multiple dynamical and thermodynamical mechanisms evolving across spatial and temporal scales.

A central mechanism underlying monsoon onset is the development of land–sea thermal contrast during the pre-monsoon months. Differential heating between the Indian landmass and surrounding oceans induces large-scale pressure gradients that drive cross-equatorial flow and strengthen low-level westerlies, particularly the Findlater (Somali) jet, which transports moisture from the Arabian Sea toward the subcontinent [23, 45]. Concurrently, the northward migration of the Intertropical Convergence Zone (ITCZ) reorganizes tropical convection, leading to enhanced cloud cover and reduced outgoing longwave radiation (OLR), both of which are widely recognized as precursors to onset [38, 62]. These processes are further modulated by large-scale pressure systems such as the Mascarene High, which influence the strength and structure of monsoon circulation.

Oceanic conditions provide an additional layer of control through their influence on atmospheric stability and energy exchange. Sea surface temperature (SST) anomalies in the Indian Ocean affect convection and moisture availability [53, 27], while mean sea level pressure (MSLP) patterns encode evolving circulation dynamics that shape large-scale moisture transport. The vertical structure of the atmosphere, captured by winds at multiple pressure levels, reflects the coupling between lower-tropospheric inflow and upper-level outflow that characterizes the mature monsoon system. Importantly, none of these variables acts in isolation; monsoon onset emerges from their nonlinear interaction across spatial scales (local to global) and temporal scales (synoptic to seasonal), making it inherently difficult to predict.

2.2 Approaches to Monsoon Onset Prediction

The challenge of predicting monsoon onset has led to a wide range of modeling approaches spanning statistical methods, dynamical models, and more recently, data-driven techniques. Statistical approaches typically rely on identifying physically meaningful predictors—such as SST, pressure gradients, winds, and radiative fluxes—and combining them through linear or low-dimensional regression models. Principal component regression (PCR), which forms the basis of operational forecasts issued by the Indian Meteorological Department (IMD), exemplifies this paradigm and achieves root mean squared errors on the order of four days at lead times of a few weeks [45]. While such models are interpretable and computationally efficient, their reliance on linear structure fundamentally limits their ability to capture nonlinear interactions among climate variables.

At the other extreme, numerical weather prediction (NWP) systems simulate the coupled evolution of the atmosphere and ocean by solving discretized physical equations. Seasonal forecasting systems such as ECMWF and related models achieve strong predictive skill and can provide forecasts at longer lead times [13]. However, these approaches require computationally intensive data assimilation pipelines and substantial domain expertise, and their outputs often require additional processing to yield precise onset predictions.

Recent advances in deep learning have introduced models capable of learning nonlinear spatio-temporal relationships directly from data. Architectures such as convolutional neural networks, ConvLSTMs, and transformer-based weather models have demonstrated strong performance in large-scale forecasting tasks [37, 47]. However, these models are typically trained in data-rich regimes and rely on large volumes of labeled data. When applied to monsoon onset prediction, where labeled samples are extremely limited, end-to-end deep learning models tend to overfit and fail to generalize reliably.

2.3 Learning in the Extreme Low-Label Regime

Monsoon onset prediction presents an extreme low-label learning problem in which each year provides a single scalar target, while the inputs consist of high-dimensional spatio-temporal climate fields. Even with decades of data, the number of labeled samples remains fewer than one hundred, creating a severe mismatch between input dimensionality and supervision. In this regime, standard end-to-end learning approaches are statistically ill-posed, as models must simultaneously learn both spatial representations and temporal decision rules from an insufficient number of observations.

This mismatch explains why increasing model capacity does not lead to improved performance, but instead exacerbates overfitting. High-capacity models attempt to fit idiosyncratic patterns in individual years rather than learning generalizable structure, leading to poor performance on unseen data. Consequently, the primary bottleneck in this problem is not model expressivity, but the ability to learn robust representations under limited supervision.

2.4 Self-Supervised Representation Learning for Climate Data

Self-supervised learning (SSL) provides a principled approach to overcoming this limitation by leveraging structure in unlabeled data. Rather than relying on scarce onset labels, SSL methods construct auxiliary objectives that enable models to learn informative representations directly from the input distribution. A prominent example is Bootstrap Your Own Latent (BYOL), which learns representations by aligning embeddings of different augmented views of the same input without requiring negative samples [24]. To prevent representation collapse and maintain informative feature diversity, such alignment objectives are often combined with variance and covariance regularization, as in VICReg [6].

For climate data, SSL is particularly advantageous because large volumes of unlabeled spatio-temporal observations are readily available. These methods can learn stable large-scale spatial structures such as circulation patterns, thermal gradients, and cloud organization without requiring labeled onset dates. An additional advantage arises from the temporal structure of the data. By decomposing spatio-temporal sequences into individual snapshots, each time step can be treated as an independent training instance, effectively expanding the dataset from N yearly samples to $N \times T$ observations. This substantially increases the effective sample size available for representation learning.

Recent work has begun to explore SSL for weather and climate applications, including pretrained encoders and vision-style objectives [39, 64]. However, much of this work focuses on representation quality in isolation and does not directly address downstream prediction under extreme label scarcity. As a result, these approaches are typically coupled with end-to-end models that still require substantial labeled data, limiting their effectiveness in this setting.

2.5 Factorized Learning for Spatio-Temporal Prediction

The limitations of existing approaches suggest that representation learning and prediction should be treated as distinct problems operating under fundamentally different data regimes. Rather than learning a direct mapping from high-dimensional inputs to the target, the problem can be factorized into two components: a spatial encoder trained using self-supervision on abundant unlabeled data, and a temporal model trained using labeled data on low-dimensional representations.

This factorization fundamentally alters the statistical structure of the problem. The representation learner operates on $N \times T$ samples, enabling it to learn stable and generalizable spatial features, while supervised learning is confined to a compact temporal model whose capacity is commensurate with the number of labeled samples. This separation acts as an implicit regularizer that reduces overfitting and improves generalization across years.

Importantly, this decomposition also aligns with the physical structure of the monsoon system. Large-scale spatial patterns evolve smoothly and can be learned independently of onset labels, while the timing of onset depends on the temporal evolution of these patterns. By modeling these two aspects separately, the factorized approach mirrors the structure of the underlying physical process rather than forcing a single model to learn both spatial abstraction and temporal decision-making from limited supervision.

Despite its conceptual appeal, prior work has not explicitly combined self-supervised spatial representation learning, temporal decomposition for sample amplification, and factorized temporal modeling in the context of monsoon onset prediction. Existing methods either rely on linear statistical models or train high-capacity neural networks end-to-end, both of which struggle under extreme data scarcity. This gap motivates the proposed framework, which leverages the complementary strengths of self-supervised learning and structured modeling to enable robust prediction in this challenging regime.

2.6 Positioning of the Present Work

The approach developed in this paper builds on these insights by explicitly decoupling spatial representation learning from temporal forecasting. By leveraging unlabeled spatio-temporal data through self-supervision and restricting supervised learning to a low-dimensional temporal model, the method addresses the fundamental mismatch between input complexity and label availability. At the same time, the use of temporal decomposition and iterative prediction aligns the learning framework with both the statistical structure of the data and the physical processes underlying monsoon onset. In doing so, the proposed framework provides a principled pathway for applying deep learning to scientific problems characterized by high dimensionality and limited supervision.

3 Problem Setup

We consider the task of predicting the Monsoon Onset over Kerala (MOK) from global atmospheric and oceanic observations during the pre-monsoon period. The goal is to learn a mapping from high-dimensional spatio-temporal climate fields to a scalar onset date for each year.

Input Representation For each year i, the input is a multivariate spatio-temporal tensor $X_i \in \mathbb{R}^{T\times H\times W\times C}$, where T denotes the number of temporal observations, (H, W) are the spatial grid dimensions, and C is the number of meteorological variables. Each time step corresponds to a global climate field, enabling the model to capture large-scale patterns relevant to monsoon evolution.

Prediction Target The target variable is the monsoon onset date declared by the India Meteorological Department (IMD). We model the target as a continuous variable relative to the climatological onset date (June 1) $y_i = \text{OnsetDate}_i - \text{June 1}$, which transforms the task into a regression problem with values centered around zero.

Dataset We use global meteorological variables from the ERA5 reanalysis dataset spanning multiple decades. Each yearly sample consists of $T = 13$ weekly observations covering the pre-monsoon period from February to May, with spatial resolution $H \times W = 481 \times 1440$ and $C = 8$ variables. The dataset contains $N = 74$ yearly samples. We use a chronological split with 1951–2010 for model development and 2011–2024 for evaluation. This setup ensures that all test samples correspond to future years unseen during training.

Input Variables The input channels correspond to key meteorological variables known to influence monsoon dynamics, including: 2-meter air temperature (T2M), capturing land–sea thermal contrast; mean sea level pressure (MSLP), representing large-scale pressure gradients; total cloud cover (TCC), indicative of convective activity; top net thermal radiation (TTR / OLR), a proxy for tropical convection; and zonal and meridional wind components at 850 hPa and 200 hPa, capturing lower and upper tropospheric circulation. Together, these variables characterize large-scale ocean–atmosphere processes that precede monsoon onset.

Learning Setting and Challenges The defining characteristic of this problem is the extreme scarcity of labeled data. While each input X_i is high-dimensional, only a single scalar target y_i is available per year, resulting in fewer than 100 labeled samples in total.

This creates a severe mismatch between input dimensionality and supervision, making the learning problem statistically ill-posed for high-capacity models. In particular:

- End-to-end supervised models are prone to overfitting due to limited labels.

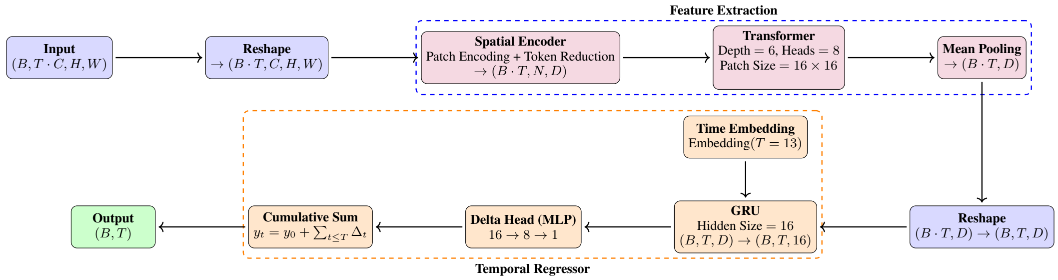

Figure 1: **Framework overview.** The pipeline factorizes the problem into spatial feature extraction and temporal forecasting. Each weekly climate snapshot is encoded independently by a shared Transformer-based spatial encoder, producing one embedding per time step, and a lightweight GRU then converts the embedding sequence into cumulative onset-date updates through the iterative refinement head.

- The input exhibits complex spatio-temporal structure with long-range dependencies across space and time.
- The underlying dynamics are nonlinear and governed by coupled ocean–atmosphere processes.

These challenges motivate the need for learning strategies that can exploit the structure of the input distribution while minimizing reliance on labeled supervision.

4 Method

4.1 Overview

The central challenge is not model capacity but statistical structure: we have fewer than 100 labeled samples, each consisting of a high-dimensional spatio-temporal tensor, and standard end-to-end learning asks a single model to jointly infer spatial representations and temporal decision rules from a sample size that can reliably constrain neither. Our approach resolves this by recognizing that the problem decomposes into two stages with fundamentally different data regimes. Spatial structure, such as large-scale circulation and thermal gradient patterns characteristic of the pre-monsoon period, can be learned from individual weekly snapshots without onset labels, whereas temporal dynamics describing how these patterns evolve toward onset require labeled years but operate in a much lower-dimensional space once representations are fixed.

We exploit this separation through a factorized framework (Figure 1) in which a shared spatial encoder ϕ maps each weekly snapshot $x_t \in \mathbb{R}^{H \times W \times C}$ to an embedding $z_t = \phi(x_t) \in \mathbb{R}^D$, trained with self-supervision across all $N \cdot T$ snapshots. A lightweight temporal model g then maps the sequence $\{z_1, \ldots, z_T\}$ to a predicted onset date $\hat{y} = g(z_1, \ldots, z_T)$, trained only on labeled years with ϕ frozen. This separation ensures that each component is trained under a data regime appropriate to its capacity.

4.2 Factorized Learning

Directly learning $f : \mathbb{R}^{T \times H \times W \times C} \to \mathbb{R}$ end-to-end forces the model to simultaneously estimate both a high-dimensional spatial encoder and a temporal predictor from N labeled samples, rendering the problem ill-posed. Factorizing $f = g \circ \phi$ fundamentally changes the statistical burden placed on each component: representation learning under self-supervision operates on $N \cdot T$ snapshot-level observations, while supervised learning is confined to a compact temporal model whose parameter count is commensurate with the available yearly labels.

A critical design choice is to freeze ϕ during supervised training. Allowing label gradients to propagate into the encoder leads to overfitting to idiosyncrasies of individual years rather than learning generalizable structure, effectively collapsing the intended separation between representation learning and temporal reasoning. Freezing the encoder therefore acts as a constraint that localizes all label-driven adaptation to the low-dimensional temporal model, which we empirically verify to be essential for generalization.

4.3 Self-Supervised Spatial Encoder

We train ϕ using a BYOL-style objective augmented with variance and covariance regularization. Each input sequence is temporally decomposed so that individual snapshots $x_{i,t}$ are treated as independent training instances, expanding the effective dataset from N to $N \cdot T$. For each snapshot, two augmented views $x^{(1)}$ and $x^{(2)}$ are generated, and the model is trained to align their representations via

$$\mathcal{L}_{\text{align}} = 2 - 2\cos\Big(p(z^{(1)}), \text{sg}(z^{(2)})\Big),$$

where $p(\cdot)$ is a predictor network and $\text{sg}(\cdot)$ denotes stop-gradient. However, alignment alone admits degenerate solutions in which all inputs collapse to a single representation, rendering the embeddings uninformative for downstream prediction.

To prevent this, we incorporate variance and covariance regularization directly into the objective. The variance term $\mathcal{L}_{\text{var}} = \sum_j \max(0, 1 - \sigma(z_j))$ ensures that each embedding dimension remains active across the batch, while the covariance term $\mathcal{L}_{\text{cov}} = \sum_{i \neq j} \text{Cov}(z_i, z_j)^2$ discourages redundancy between dimensions. The full objective is therefore

$$\mathcal{L}_{\text{SSL}} = \mathcal{L}_{\text{align}} + \lambda_1 \mathcal{L}_{\text{var}} + \lambda_2 \mathcal{L}_{\text{cov}},$$

and we find that both regularization terms are necessary to obtain representations that are both stable and predictive. Appendix C covers this in more detail.

4.4 Temporal Modeling via Iterative Refinement

With spatial embeddings fixed, the temporal model operates on the sequence $\{z_1, \ldots, z_T\}$ and produces an onset prediction through iterative refinement rather than a single terminal estimate. Specifically, the model predicts incremental updates Δ_t at each timestep and accumulates them into a running forecast $\hat{y}_t = y_0 + \sum_{i=1}^{t} \Delta_i$, where y_0 is a learnable scalar initialized to the training mean. This formulation enables the model to produce calibrated predictions at any point in the sequence, which is operationally valuable, while also imposing a structural bias toward smooth temporal evolution consistent with the gradual development of large-scale monsoon precursors.

In practice, a GRU processes the embedding sequence to produce hidden states, which are combined with learnable time embeddings and passed through a lightweight feedforward head to predict Δ_t. This design keeps the temporal model expressive enough to capture non-linear dynamics while maintaining a parameter count appropriate for the limited number of labeled years.

4.5 Training

Training proceeds in two stages. First, the spatial encoder ϕ is trained using $\mathcal{L}_{\text{SSL}}$ on all $N \cdot T$ unlabeled snapshots. In the second stage, ϕ is frozen and the temporal model is trained on labeled years using a composite loss

$$\mathcal{L} = \frac{(\hat{y}_T - y)^2 + \alpha \mathcal{L}_{\text{time}} + \beta \mathcal{L}_{\text{smooth}} + \gamma \mathcal{L}_{\text{mono}}}{1 + \alpha + \beta + \gamma},$$

where the primary term supervises the final prediction, $\mathcal{L}_{\text{time}}$ provides time-weighted supervision to intermediate predictions, $\mathcal{L}_{\text{smooth}}$ penalizes large updates to regularize the refinement trajectory, and $\mathcal{L}_{\text{mono}}$ discourages prediction error from increasing over time. Hyperparameters are selected via grid search on a validation set, with full details provided in Appendix D.

5 Experiments

5.1 Setup

Evaluation protocols. We evaluate models under two protocols. The *fixed split* trains on 1951–1970 and 1981–2010, uses the split 1971-1980 for validation, and evaluates on 2011–2024, providing a clean comparison between a fully trained model and held-out future years. The *leave-one-year-out (LOYO)* protocol trains separately for each test year i in 2011-2024 on all years preceding i n the dataset, except 1971-1980 which are used for validation, and evaluates on year i. LOYO enforces

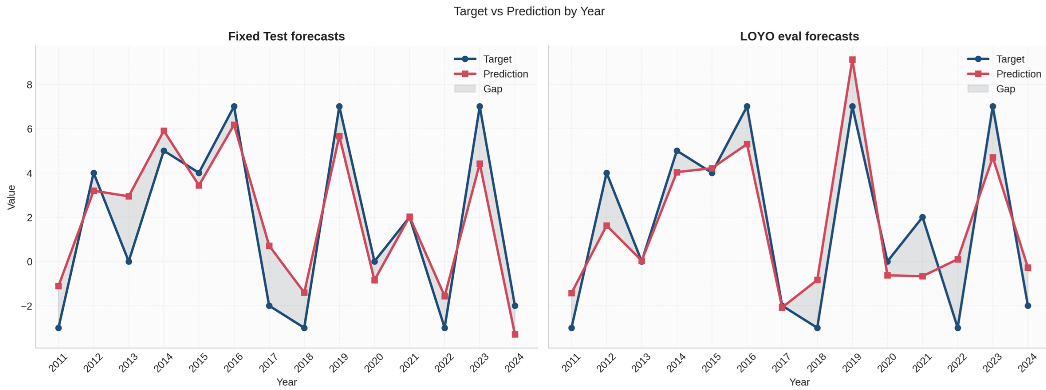

Figure 2: Predicted versus true monsoon onset dates for the held-out years under the fixed chronological split and the leave-one-year-out (LOYO) protocol. The figure shows that the factorized model tracks the observed onset anomalies closely in both settings, with the LOYO predictions remaining well aligned despite the stricter temporal generalization constraint.

Setting	Method	RMSE ↓	Skill ↑
Fixed Train–Test	Climatology	5.16	0.00
	PCR	4.00	0.40
	CNN + MLP	3.93	0.43
	ConvLSTM	4.86	0.12
	ViT	4.81	0.15
	Ours	**1.64**	**0.90**
LOYO	Climatology	4.98	0.00
	Ours	**1.81**	**0.87**

Table 1: Performance comparison under the fixed chronological train–test split and the more stringent LOYO evaluation. The table reports RMSE in days together with skill relative to climatology, highlighting that the proposed factorized model outperforms both statistical baselines and end-to-end deep learning models.

strict temporal causality, i.e., the model for year i has access only to data from years before i and closely simulates operational deployment in which predictions must be issued before the target event occurs. Performance is reported as RMSE in days and as skill score relative to climatology, $S = 1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{clim}}$, where $S = 0$ indicates no improvement over the seasonal mean and $S = 1$ indicates perfect prediction.

Baselines We compare against representative approaches across modeling paradigms. Statistical baselines include climatology and principal component regression (PCR). Deep learning baselines include end-to-end models operating directly on spatio-temporal inputs like CNN, ConvLSTM, and ViT, which must learn representations and temporal dynamics jointly. To isolate the contribution of our method, we also evaluate feature ablations where the pipeline is trained end-to-end, where trained encoder is fine-tuned instead of frozen, and testing random features to show that our features are meaningful, and temporal regressor does not fit to noise. We also perform ablations on the SSL objective by comparing it against ViT-MAE reconstruction task, and BYOL without regularization. Across all these ablations, the temporal model design is kept constant.

5.2 Main Results

Table 1 establishes three consistent patterns. First, the factorized model does not just marginally improve over the baselines; the margin is decisive. RMSE drops from 4.00 for PCR and 3.93 for the strongest end-to-end deep baseline (CNN + MLP) to 1.64 days, a reduction of more than 50% relative to the operational statistical baseline. Relative to climatology, this corresponds to a skill score of 0.90, indicating that the model accounts for nearly all of the residual variance that remains

Year	Ground Truth	Fixed Test Pred.	LOYO Pred.
2011	-3	-1.11	-0.51
2012	4	3.19	5.04
2013	0	2.94	2.72
2014	5	5.90	6.98
2015	4	3.43	5.49
2016	7	6.17	5.27
2017	-2	0.71	0.34
2018	-3	-1.41	-1.66
2019	7	5.66	6.38
2020	0	-0.85	2.53
2021	2	2.02	0.15
2022	-3	-1.57	-1.24
2023	7	4.42	5.10
2024	-2	-3.29	-1.49

Table 2: Year-wise comparison between the observed onset anomaly and the predictions produced by the fixed-split and LOYO models for each test year. The table makes the temporal variation explicit and shows that the model remains accurate across both near-neutral and strongly shifted onset years.

after a persistence-style predictor. To put this concretely: the Indian Meteorological Department's operational PCR method achieves an RMSE of 4 days at a lead time of approximately 15 days [45]; our method achieves 1.64 days at a lead time of approximately 30 days, with predictions issued from as early as 1 February.

Second, the comparison with deep learning baselines is diagnostic. ConvLSTM (RMSE 4.86) and ViT (RMSE 4.81) both underperform PCR, which confirms that greater architectural capacity does not compensate for insufficient supervision. These models must learn both the feature extractor and the temporal decision rule from fewer than 100 labeled years, and the results make clear that joint optimization is too unstable in this regime. The factorized design sidesteps this problem entirely by decoupling the two objectives, and the performance gap reflects that structural difference rather than any incidental advantage.

Third, the gap between the fixed-split and LOYO evaluations is remarkably small. RMSE increases only from 1.64 to 1.81 days, and skill remains high at 0.87 under the stricter protocol. That modest degradation is informative: it shows the method is not memorizing the holdout period but learning a year-agnostic mapping that transfers to future onset years it has never seen.

The year-wise results in Table 2 reinforce this interpretation. Predictions remain tightly aligned with the observed anomalies across a wide range of onset conditions, including near-neutral years such as 2013 (target: 0, fixed prediction: 2.94) and years with strong deviations such as 2016 (target: 7, fixed prediction: 6.17) and 2018 (target: -3, fixed prediction: -1.41). Crucially, the model captures both the sign and magnitude of the anomaly consistently, not just the central tendency. This consistency across diverse onset regimes is a stronger signal than average RMSE alone, because it indicates the model preserves ordering and scale across conditions that differ substantially from the training mean.

Overall, the main result is that factorization transforms the effective learning problem. Instead of asking a single high-capacity network to jointly infer representations and temporal dynamics from scarce labels, the proposed method learns stable spatial features under abundant self-supervision and then fits a compact temporal regressor. The gains in Table 1 are therefore best understood as a consequence of this structural alignment with the data regime, not as an incidental architectural effect.

5.3 Early Prediction from Partial Sequences

A practically important question is how far in advance the model can provide useful guidance. Figure 3 and Table 3 address this directly by evaluating forecast quality as a function of the available input horizon T'.

The results are striking in two respects. First, the model begins to outperform climatology well before the sequence is complete. In the fixed-split setting, meaningful skill ($S > 0$) emerges at $T' = 4$,

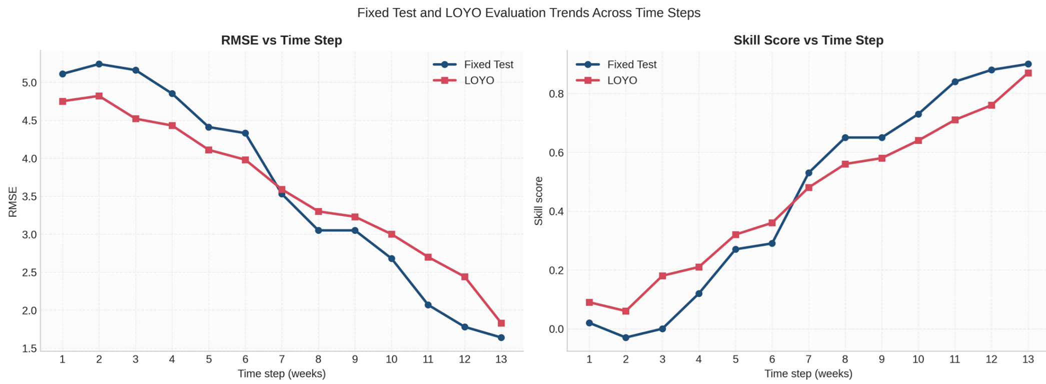

Figure 3: Forecast performance as a function of input horizon T'. The model begins to outperform climatology around $T' = 4$ and continues to improve as more observations are incorporated, demonstrating effective utilization of early signals.

	Fixed		**LOYO**	
T′	**Skill ↑**	**RMSE ↓**	**Skill ↑**	**RMSE ↓**
1	0.02	5.11	0.07	4.78
2	-0.03	5.24	0.07	4.79
3	0.00	5.16	0.13	4.64
4	0.12	4.85	0.32	4.11
5	0.27	4.41	0.51	3.49
6	0.29	4.33	0.61	3.09
7	0.53	3.53	0.68	2.80
8	0.65	3.05	0.71	2.64
9	0.65	3.05	0.72	2.62
10	0.73	2.68	0.82	2.11
11	0.84	2.07	0.85	1.90
12	0.88	1.78	0.86	1.86
13	0.90	1.64	0.87	1.81

Table 3: Forecast performance as a function of the available input horizon T', measured by RMSE and skill for both the fixed split and LOYO settings. This table complements the early-forecast figure by showing how predictive accuracy improves as additional weekly observations are revealed.

corresponding to data available through early March, roughly three months before the climatological onset date. Under the more demanding LOYO protocol, skill at $T' = 4$ is already 0.32, with RMSE reduced to 4.11 from the climatological baseline of 4.98.

Second, performance improves steadily and monotonically as more observations are incorporated. RMSE falls from 5.11 at $T' = 1$ to 1.64 at $T' = 13$ in the fixed split, and from 4.78 to 1.81 under LOYO, without any late-stage collapse. The largest gains are concentrated between $T' = 4$ and $T' = 8$, where RMSE drops from 4.85 to 3.05 in the fixed split and from 4.11 to 2.64 in LOYO. This suggests that the most discriminative onset signals are present in the early-to-mid pre-monsoon period, broadly consistent with the known importance of February-to-April circulation patterns for monsoon predictability.

The shape of the learning curve reflects the design of the iterative refinement formulation. Rather than waiting for the final timestep and producing a single high-variance prediction, the model updates its estimate incrementally as evidence accumulates. The near-monotonic decrease in RMSE across all timesteps confirms that the refinement mechanism is stable: each additional weekly observation provides a consistent marginal improvement, and the model never deteriorates as the sequence grows.

From an operational standpoint, these results have direct implications. At $T' = 10$ (data available through late April), RMSE is already 2.68 in the fixed split and 2.11 under LOYO, with skill scores of

Component	Variant	RMSE (days) ↓
Representation	Random features	3.18 ± 0.23
	Learned features (Ours)	**1.64**
Training Strategy	End-to-end training	4.14
	Fine-tuning encoder	4.03
	Frozen encoder (Ours)	**1.64**
SSL Objective	ViT-MAE	3.26
	BYOL (no reg.)	3.63
	BYOL + reg. (Ours)	**1.64**

Table 4: Ablation study for the representation-learning pipeline. The table isolates the effect of learned versus random features, frozen versus fine-tuned spatial encoders, and different self-supervised objectives, showing that the combination of BYOL with regularization and a frozen encoder is critical in the low-label regime.

0.73 and 0.82, respectively. This places the model well ahead of the IMD's operational PCR forecast issued around 1–15 May, and with longer lead time. The model therefore supports a meaningful advance in practical forecast capability, delivering skillful predictions across a range of lead times rather than only at the end of the pre-monsoon window.

5.4 Ablation Study

Table 4 isolates the contribution of each component of the proposed design and confirms that the performance gains do not arise from any single element in isolation.

The first comparison concerns the learned representation itself. Replacing the trained encoder with random features increases RMSE from 1.64 to 3.18 ± 0.23 days. This gap establishes that the spatial encoder captures physically meaningful structure, not merely a generic compression of the input. The variance across seeds for the random-feature baseline also indicates that the temporal regressor, on its own, is unable to compensate for uninformative inputs, even under favorable random projections.

The second comparison isolates the effect of training strategy. End-to-end training (RMSE 4.14) and encoder fine-tuning (RMSE 4.03) both substantially underperform the frozen-encoder configuration, despite having the same underlying architecture. This result has a clear interpretation: the temporal regressor is straightforward to train once the representation is held fixed, but the encoder becomes fragile under gradient updates from only a handful of yearly labels. Freezing the encoder is therefore not merely a regularization choice; it is the mechanism by which the method avoids overfitting the representation to spurious year-specific patterns.

The third comparison examines the choice of self-supervised objective. ViT-MAE (RMSE 3.26) improves over random features but falls well short of the proposed approach, suggesting that reconstruction-based pretraining does not produce the kind of compact, invariant representations that support downstream forecasting in this regime. BYOL without regularization (RMSE 3.63) degrades further still, indicating that the alignment objective alone is insufficient. The variance and covariance regularization terms are necessary to prevent embedding collapse and maintain the diversity of the learned representations, without which the downstream temporal model cannot distinguish among different onset regimes.

Taken together, these ablations identify three conditions that are jointly necessary for strong performance: (i) a learned spatial encoder that captures stable, physically relevant structure; (ii) a frozen encoder during supervised training, which restricts label consumption to the low-dimensional temporal model; and (iii) a regularized BYOL objective that preserves representation diversity. The removal of any single component leads to a substantial degradation. This tight interdependence underlines that the proposed method works as a system, and that its effectiveness in the low-label regime is a consequence of the overall design rather than any individual component.

6 Discussion

6.1 Data Scarcity as the Binding Constraint

A natural question is whether the gains shown in Table 1 could be recovered by a better-designed end-to-end model, or whether data scarcity is genuinely the binding constraint. The ablation evidence argues strongly for the latter. The problem here is not that we lack labels for existing inputs; it is that each yearly sample is a single observation of a slow-timescale process, and 74 years of data is all that exists. No amount of architectural sophistication changes this: an end-to-end model, however well designed, must still estimate its full parameter stack from 60 training years.

This is why end-to-end training with the same spatial encoder degrades to 4.14 days RMSE, and fine-tuning it degrades to 4.03, both far worse than the frozen-encoder result despite access to identical pretrained features. The problem is not the quality of the starting representation but what happens to it under gradient updates from so few samples. High-dimensional encoders have far more parameters than 60 yearly observations can reliably constrain, and any fine-tuning in this regime fits the encoder to idiosyncrasies of the training years rather than to generalizable structure. This is also why capacity alone, as demonstrated by ConvLSTM and ViT, does not help: the issue is one of sample size, not expressiveness.

Factorization addresses this directly by changing what the yearly data is asked to do. The self-supervised stage exploits temporal decomposition to operate on approximately 960 snapshot-level observations rather than 74 yearly ones, learning stable spatial structure with a variance commensurate with that sample size. The supervised stage then only needs to fit a compact temporal model whose parameter count is commensurate with the available years. The performance gains are a consequence of this alignment between model complexity and data availability at each stage.

6.2 Interpreting the Early Prediction Curve

The near-monotonic improvement of RMSE across $T' = 1$ to $T' = 13$ has a specific interpretation. It means predictive information about the onset date is not localized to a narrow window near the end of the pre-monsoon sequence, but distributed broadly across weeks. This is consistent with the known dynamics of large-scale monsoon precursors, which evolve gradually over months rather than switching abruptly in the days before onset. A model that waited until late April to issue a prediction would be discarding most of this signal.

The gain concentration between $T' = 4$ and $T' = 8$ is an empirical observation that warrants further investigation. This window corresponds to March through mid-April, and the known importance of early-to-mid pre-monsoon signals such as Findlater jet intensification and Arabian Sea SST evolution in the dynamical literature offers a plausible explanation.

6.3 Limitations

The assumption that spatial representations are time-invariant is a deliberate simplification. In reality, the spatial structure of relevant climate fields does evolve across the pre-monsoon sequence, and a shared encoder trained on snapshots pooled across all timesteps cannot capture this evolution at the representation level. The temporal model compensates by operating on the sequence of embeddings, but interactions that require joint spatial-temporal reasoning may be lost at the factorization boundary.

The evaluation covers 14 test years. While the LOYO protocol is causally strict and the results are internally consistent, the test set is small, and performance on years with genuinely exceptional dynamics may not be fully representative of the method's behavior at the tails of the onset distribution.

Finally, the self-supervised representations are learned from the historical ERA5 record, which reflects a climate that is itself changing. If pre-monsoon circulation patterns shift systematically outside the range of variability seen during 1951–2024, the learned embeddings may no longer capture the features most relevant to onset prediction. This is not a failure specific to our method, but it is a limitation worth acknowledging given the long intended deployment horizon of operational forecasting systems.

7 Conclusion

Monsoon onset prediction is a hard instance of a general problem: the inputs are high-dimensional and structured, but the process being predicted unfolds once per year, and decades of observations are all that exist. Rather than accepting this as a fixed constraint, we showed that the problem can be restructured. By decoupling spatial representation learning from temporal forecasting, the self-supervised stage operates on nearly 1000 snapshot-level observations while the supervised stage operates on a compact model whose complexity is commensurate with the available yearly samples. This alignment between model structure and data availability at each stage is the reason the method works, and the ablation results confirm it: each component of the factorized design is necessary, and the gains collapse when any one of them is removed.

The practical implications are significant. At roughly twice the lead time of the IMD's operational forecast, the proposed method reduces prediction error by more than half. Skillful predictions are available from early March onward, well before the pre-monsoon sequence is complete, and the forecast improves stably week by week through late April. This is not just a benchmark improvement; it represents a qualitatively different kind of forecast, one that can support earlier and more confident decision-making in water management and agriculture.

More broadly, the factorized framework is not specific to monsoon prediction. The structural conditions it exploits, abundant unlabeled spatio-temporal observations of a process that can only be observed at coarse temporal resolution, arise across many scientific domains. The principle of reserving yearly observations for a compact model operating in a learned embedding space is general, and we expect it to transfer to other settings where the slow timescale of the target process makes end-to-end learning intractable.

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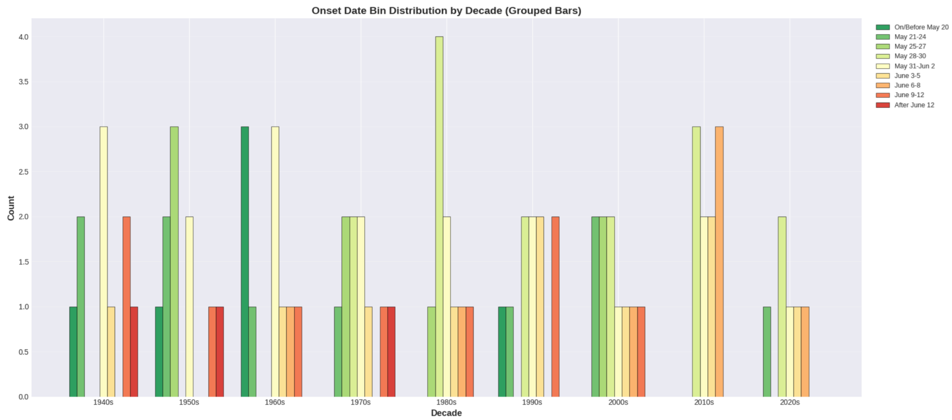

Figure 4: Distribution of monsoon onset targets across the full year range used in the study. This figure summarizes the spread of onset anomalies and shows that the evaluation set covers both early and late onset years rather than a narrow slice of the target distribution.

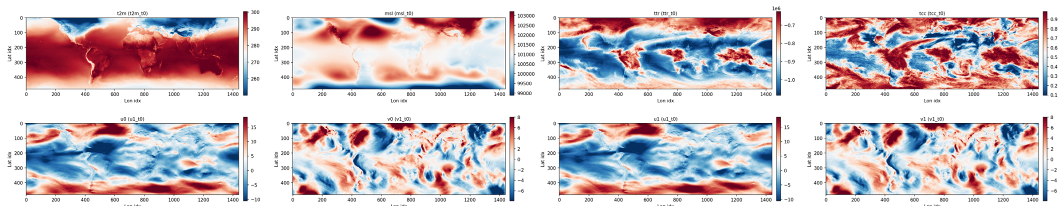

Figure 5: Representative ERA5 input fields used by the model. The visualization illustrates the type of multi-variable, global spatio-temporal climate snapshots that are encoded at each weekly timestep before being passed to the self-supervised spatial encoder.

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A Dataset Details

A.1 Data Source

We use the ERA5 reanalysis dataset provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 combines observations from satellites, weather stations, and other sources through a data assimilation system to produce globally consistent atmospheric and oceanic fields. The data is available on a regular latitude–longitude grid at a spatial resolution of $0.25^\circ \times 0.25^\circ$.

A.2 Temporal Coverage and Splits

The dataset spans the period 1951–2024, yielding a total of 74 yearly samples. Each sample corresponds to a single year with one associated target value (monsoon onset date).

We use the following chronological splits:

- **Training set:** 1951–1970 + 1981-2010
- **Validation set:** 1971–1980
- **Test set:** 2011–2024

Chronological splitting is used to prevent information leakage from future years into training and validation. The validation subset is chosen as a contiguous subset that best represents the range of the train set.

In addition to the fixed split, we also evaluate using a leave-one-year-out (LOYO) protocol. For each test year i, the model is trained on all years prior to i and evaluated on year i, with validation performed on a fixed validation subset.

A.3 Input Representation

For each year, the input consists of multivariate spatio-temporal climate fields represented as a tensor:

$$X \in \mathbb{R}^{T\times H\times W\times C}, \tag{1}$$

where:

- $T = 13$ is the number of temporal observations,
- $H = 481$ and $W = 1440$ are the spatial grid dimensions,
- $C = 8$ is the number of meteorological variables.

The temporal dimension corresponds to weekly observations spanning the pre-monsoon period (February to May). Weekly snapshots are obtained by averaging daily ERA5 fields over non-overlapping 7-day windows.

The spatial domain covers the global latitude–longitude grid.

A.4 Meteorological Variables

We use eight physically relevant meteorological variables known to influence monsoon dynamics. These variables capture large-scale oceanic and atmospheric processes such as thermal gradients, circulation patterns, and convection. Table A.4 represents these 8 variables.

Variable	**Symbol**	**Units**	**Description**
2m Air Temperature	T2M	K	Land–sea thermal contrast
Mean Sea Level Pressure	MSLP	Pa	Large-scale pressure gradients
Total Cloud Cover	TCC	Fraction	Convective activity proxy
Top Net Thermal Radiation (OLR)	TTR/OLR	W/m^2	Tropical convection proxy
U-Wind (850 hPa)	U1	m/s	Lower tropospheric circulation
V-Wind (850 hPa)	V1	m/s	Lower tropospheric circulation
U-Wind (200 hPa)	U0	m/s	Upper tropospheric circulation
V-Wind (200 hPa)	V0	m/s	Upper tropospheric circulation

Table 5: The eight meteorological variables used as input channels, along with their symbols, physical units, and the monsoon-related phenomenon each variable helps capture. These variables span thermal, pressure, cloud, radiation, and wind fields that collectively describe pre-monsoon evolution.

A.5 Target Variable

The prediction target is the monsoon onset date over Kerala (MOK) as defined by the objective criteria of the India Meteorological Department (IMD). Instead of predicting the absolute calendar date, we express the target as a deviation from the climatological onset date (June 1):

$$y = \text{Onset Date} - \text{June } 1. \tag{2}$$

This formulation converts the task into a continuous regression problem. The target values are approximately centered around zero, with observed values ranging from -19 to $+17$ days.

A.6 Effective Sample Size

A key challenge of this problem is the extremely limited number of labeled samples, as only one onset date is available per year (74 samples total).

To mitigate this, we exploit temporal structure during representation learning by treating individual time steps as independent samples. This increases the effective number of training samples for the self-supervised stage from 74 to:

$$74 \times 13 = 962. \tag{3}$$

This temporal decomposition is used only during representation learning; supervised training is still performed at the yearly level.

A.7 Spatial Coverage

The input data consists of global climate fields, allowing the model to capture large-scale teleconnections and ocean–atmosphere interactions that influence monsoon onset. No spatial masking is applied, and all grid locations are used during training.

A.8 Data Quality and Missing Values

ERA5 is a reanalysis dataset with complete spatial and temporal coverage. No missing values are present in the extracted variables, and no imputation is required.

A.9 Data Leakage Prevention

To ensure a fair evaluation, all preprocessing statistics (e.g., climatological means and normalization parameters) are computed exclusively using the training data. These statistics are then applied to validation and test sets without modification, preventing any leakage of future information.

B Preprocessing

B.1 Overview

We apply a spatially varying normalization procedure to the ERA5 variables to ensure numerical stability while preserving spatial structure. All preprocessing statistics are computed exclusively from the training data.

B.2 Spatially and Temporally Varying Normalization

The input tensor is reshaped such that temporal and channel dimensions are combined into a single channel dimension. Specifically, an input

$$X \in \mathbb{R}^{T \times H \times W \times C}$$

is reshaped to:

$$X \in \mathbb{R}^{(T \cdot C) \times H \times W}.$$

As a result, each pair of variable and time step is treated as a distinct channel during preprocessing.

For each combined channel $k \in \{1, \ldots, T \cdot C\}$ and spatial location (h, w), we compute the mean and standard deviation over all training samples:

$$\mu_k(h, w) = \frac{1}{N_{\text{train}}} \sum_{i \in \mathcal{D}_{\text{train}}} X_{i,k,h,w}, \tag{4}$$

$$\sigma_k(h, w) = \sqrt{\frac{1}{N_{\text{train}} - 1} \sum_{i \in \mathcal{D}_{\text{train}}} (X_{i,k,h,w} - \mu_k(h, w))^2}. \tag{5}$$

The normalized input is:

$$\tilde{X}_{i,k,h,w} = \frac{X_{i,k,h,w} - \mu_k(h, w)}{\sigma_k(h, w) + \epsilon}, \tag{6}$$

where $\epsilon = 10^{-6}$.

This formulation allows normalization statistics to vary across both spatial locations and temporal indices, effectively capturing season-specific distributions for each variable.

B.3 Interpretation

By treating each time step as a separate channel, the normalization process implicitly accounts for seasonal variation in climate variables. This avoids enforcing a shared distribution across different phases of the pre-monsoon period, which may exhibit distinct statistical properties.

B.4 Handling Low-Variance Regions

For grid locations where the standard deviation is extremely small ($\sigma_c(h, w) < 10^{-8}$), we set $\sigma_c(h, w) = 1$ to avoid numerical instability.

B.5 Order of Operations

The preprocessing pipeline is applied in the following order:

1. Extract raw ERA5 variables
2. Aggregate daily data into weekly observations
3. Compute mean and standard deviation using training data
4. Apply spatially varying normalization
5. Apply SST masking using land–sea mask

B.6 Data Leakage Prevention

All normalization statistics are computed exclusively using the training set. These statistics are reused for validation and test data without recomputation.

B.7 Implementation Details

Normalization statistics are computed once and cached for reuse across experiments. The statistics have shape (C, H, W), allowing location-specific normalization.

C Self-Supervised Representation Learning

C.1 Overview

We learn spatial representations using a self-supervised learning (SSL) framework based on Bootstrap Your Own Latent (BYOL), augmented with additional regularization to prevent representation collapse.

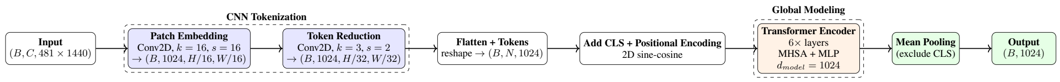

Figure 6: Spatial encoder architecture used during self-supervised pretraining. The model first tokenizes each global climate field with convolutional patch embedding and token reduction, then applies a Transformer with fixed positional encodings, and finally mean-pools the token sequence into a single latent vector per timestep.

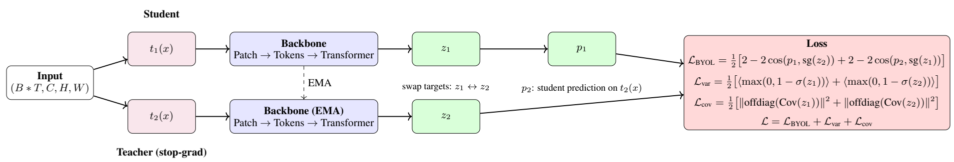

Figure 7: Self-supervised BYOL training setup with two augmented views of the same climate snapshot. A student network predicts the representation produced by an EMA teacher, while variance and covariance regularizers discourage representational collapse and encourage diverse features across the embedding dimensions.

C.2 Temporal Flattening

To increase the effective number of training samples, we treat each temporal snapshot as an independent sample. An input tensor

$$X \in \mathbb{R}^{B \times T \times C \times H \times W}$$

is reshaped to:

$$X \in \mathbb{R}^{(B \cdot T) \times C \times H \times W}.$$

C.3 Model Architecture

The SSL model consists of:

- A Transformer-based spatial encoder with:
  - embedding dimension = 1024
  - depth = 6 layers
  - number of heads = 8
- A projection head of dimension 1024
- A prediction head with output dimension 512

C.4 BYOL Objective

Given two augmented views x_1 and x_2, the model computes:

$$p_1 = q(g(f_\theta(x_1))), \tag{7}$$
$$z_2 = g(f_\xi(x_2)), \tag{8}$$

with the BYOL loss:

$$\mathcal{L}_{\text{BYOL}} = 2 - 2 \cdot \frac{\langle p_1, z_2 \rangle}{\|p_1\| \|z_2\|}. \tag{9}$$

The loss is symmetrized across both directions.

C.5 Anti-Collapse Regularization

We incorporate additional regularization terms:

Variance Regularization

$$\mathcal{L}_{\text{var}} = \frac{1}{D} \sum_{d=1}^{D} \max(0, 1 - \sigma_d) \tag{10}$$

Covariance Regularization

$$\mathcal{L}_{\text{cov}} = \frac{1}{D^2} \sum_{i \neq j} \text{Cov}(z_i, z_j)^2 \tag{11}$$

Final Loss

$$\mathcal{L} = \mathcal{L}_{\text{BYOL}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cov}}, \tag{12}$$

with $\lambda_{\text{var}} = 1$ and $\lambda_{\text{cov}} = 1$. These losses are symmetrized across both directions.

C.6 Momentum Encoder

The teacher encoder is updated using exponential moving average:

$$\xi \leftarrow m\xi + (1 - m)\theta, \tag{13}$$

where the momentum coefficient m is annealed from 0.9 to 0.999 using a cosine schedule.

C.7 Data Augmentation

Two independent augmented views are generated using the following transformations:

- Gaussian noise with standard deviation 0.1
- Spatial masking with probability 0.4
- Gaussian blur with probability 0.3 and $\sigma = 0.1$
- Random cropping with probability 0.5 and scale range $[0.6, 1.0]$

All augmentations are applied independently to each view.

C.8 Optimization Details

We train the model using the AdamW optimizer with:

- learning rate = 10^{-4}
- weight decay = 10^{-6}
- cosine annealing learning rate schedule

Training is performed for up to 100 epochs, and the best model is selected based on a validation score combining alignment and embedding diversity.

C.9 Implementation Details

Training is performed with deterministic initialization and data loading to ensure reproducibility. Embedding collapse is monitored during training using statistics of embedding variance and cosine similarity.

D Temporal Sequence Modeling

D.1 Overview

Given per-timestep feature representations extracted via self-supervised learning, we model monsoon onset prediction as a sequential refinement problem. Instead of directly predicting the final onset date, we predict a sequence of intermediate estimates that progressively converge to the final target.

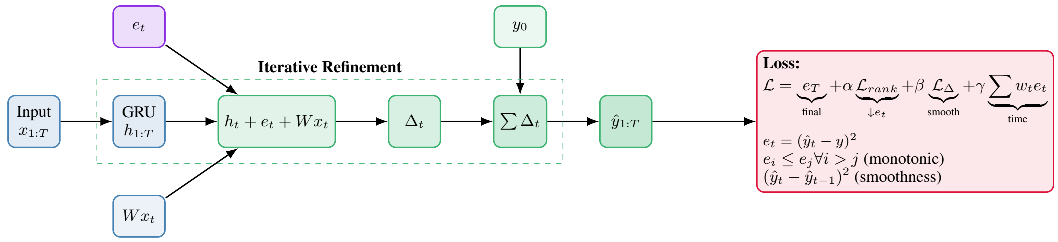

Figure 8: Temporal regressor architecture for iterative refinement. GRU hidden states are combined with time embeddings and projected input features to predict stepwise deltas Δ_t, which are accumulated into a running onset forecast and constrained by monotonicity and smoothness terms.

D.2 Delta-Based Sequence Modeling

We parameterize the prediction as a cumulative sum of learned increments:

$$y_t = y_0 + \sum_{i=1}^{t} \Delta_i, \tag{14}$$

where Δ_i denotes the predicted increment at timestep i, and y_0 is a learnable initial prediction initialized to the mean target value over the training set.

This formulation encourages smooth temporal evolution and provides an inductive bias for progressive refinement.

D.3 Model Architecture

Given input features $X \in \mathbb{R}^{B \times T \times F}$, the model consists of:

- A GRU encoder with hidden dimension 16 and 1 layer
- A learnable time embedding added to hidden states
- A residual projection of the input features
- A feedforward head predicting Δ_t at each timestep

The predicted sequence is obtained via cumulative summation of Δ_t.

D.4 Loss Function

We train the model using a composite loss that enforces accuracy, temporal consistency, and smoothness.

Final Prediction Loss

$$\mathcal{L}_{\text{final}} = \text{MSE}(y_T, y) \tag{15}$$

Time-Weighted Supervision

$$\mathcal{L}_{\text{time}} = \frac{1}{T} \sum_{t=1}^{T} w_t (y_t - y)^2 \tag{16}$$

where weights w_t increase linearly with time.

Monotonic Error Constraint

$$\mathcal{L}_{\text{rank}} = \frac{1}{T^2} \sum_{i<j} \max(0, (e_j - e_i)), \tag{17}$$

where $e_t = (y_t - y)^2$, encouraging prediction error to decrease over time.

Delta Smoothness

$$\mathcal{L}_{\text{delta}} = \frac{1}{T}\sum_{t=2}^{T} w_t(\Delta_t)^2 \tag{18}$$

Total Loss

$$\mathcal{L} = \frac{\mathcal{L}_{\text{final}} + \alpha\mathcal{L}_{\text{rank}} + \beta\mathcal{L}_{\text{delta}} + \gamma\mathcal{L}_{\text{time}}}{1+\alpha+\beta+\gamma} \tag{19}$$

D.5 Training Details

We train the model using the AdamW optimizer with:

- learning rate = 10^{-3}
- batch size = 10
- maximum epochs = 300
- early stopping patience = 20

Hyperparameters are selected via grid search over:

- $\alpha \in \{1.3, 1.5, 1.7\}$
- $\beta \in \{0.5, 1.0, 1.5\}$
- $\gamma \in \{1.5, 1.7, 1.9\}$

We evaluate each configuration across multiple random seeds $\{51, 53, 55, 57\}$ and select the model with the best validation performance. The best hyper-parameters were found to be $\alpha = 1.3, \beta = 1.0, \gamma = 1.9,$ best seed $= 55$.

D.6 Evaluation Protocol

Model selection is based on validation mean squared error (MSE) at the final timestep. During evaluation, only the final prediction y_T is used, while intermediate predictions are used solely for training.

E Sequential Leave-One-Year-Out Evaluation

E.1 Overview

To evaluate model performance in a realistic forecasting setting, we adopt a sequential leave-one-year-out (LOYO) protocol. Unlike standard LOYO evaluation, this setup simulates a real-world scenario in which models are retrained as new observations become available over time.

E.2 Evaluation Protocol

Let the test years be ordered chronologically as $\{t_1, t_2, \ldots, t_N\}$. For each test year t_i, we construct an augmented training set as:

$$\mathcal{D}_{\text{train}}^{(i)} = \mathcal{D}_{\text{train}} \cup \{t_1, t_2, \ldots, t_{i-1}\}. \tag{20}$$

At each iteration i:

1. A model is trained from scratch using $\mathcal{D}_{\text{train}}^{(i)}$
2. A fixed validation set is used for model selection
3. The trained model is evaluated only on the current test year t_i

This procedure is repeated sequentially for all test years without shuffling.

E.3 Causality and Data Leakage

The evaluation protocol is strictly causal:

- Only past years $\{t_1, \ldots, t_{i-1}\}$ are used when predicting t_i
- No information from future test years is used at any stage

Thus, the setup avoids data leakage and closely mirrors real-world deployment, where models are updated as new yearly observations become available.

E.4 Model Training Per Iteration

For each test year t_i, the model is retrained from scratch using the augmented training set. The same architecture and training procedure are used across all iterations.

Training is performed using:

- AdamW optimizer with learning rate 10^{-3}
- Batch size = 10
- Maximum of 300 epochs
- Early stopping with patience = 20 based on validation MSE

E.5 Hyperparameter Selection

Hyperparameters are selected via a discrete grid search over the loss weights:

- $\alpha \in \{0.6, 1.0, 1.4\}$ (ranking loss weight)
- $\beta \in \{0.6, 1.1, 1.6\}$ (delta smoothness weight)
- $\gamma \in \{1.6, 1.8, 2.0\}$ (time-weighted supervision weight)

This results in a total of 27 configurations. The search space is defined *a priori* and is not adapted based on test performance.

For each configuration, models are trained using multiple random seeds $\{51, 53, 55, 57\}$ to account for stochasticity. The configuration achieving the lowest validation MSE is selected and used for evaluation. The best hyper-parameters were found to be $\alpha = 0.6, \beta = 0.6, \gamma = 2.0$.

E.6 Seed Selection Per Year

For each test year t_i, multiple models corresponding to different random seeds are trained. The model achieving the lowest validation MSE is selected for that year and used for prediction.

E.7 Implementation Details

At each iteration:

- The model is initialized from scratch
- The initial prediction y_0 is set to the mean target value of the current training set
- Training and validation splits remain consistent across iterations

Predictions are generated only for the final timestep y_T, while intermediate predictions are used solely for training.

F Ablation Studies

We perform a comprehensive set of ablation studies to evaluate the impact of (i) spatial encoder hyper-parameter choices, and (ii) choice of temporal model. All results are reported using RMSE (lower is better).

Configuration	RMSE (days)
No Positional Encoding	3.64
Learnable Positional Encoding	2.81
Fixed Positional Encoding (Ours)	**1.64**
Patch Size = 8	3.03
Patch Size = 16 (Ours)	**1.64**
Patch Size = 32	2.71
No Reduction	3.77
Stride 4 Reduction	3.10
Stride 2 Reduction (Ours)	**1.64**

Table 6: Ablation study of the spatial encoder design. The table compares positional encoding choices, patch sizes, and token-reduction strides, showing how each architectural choice affects the downstream onset RMSE after factorized training.

Model	RMSE
MLP	3.03
Random Forest	3.49
XGBoost	2.90
LSTM + Refinement	2.58
Attention + Refinement	2.50
GRU + refinement (Ours)	**1.64**

Table 7: Comparison of temporal forecasting models used on top of the learned spatial embeddings. The table shows that simple regressors, classical machine-learning methods, and alternative sequence models all lag behind the GRU-based refinement module adopted in the proposed framework.

The results in Tables 6 and 7 demonstrate that performance is highly sensitive to both spatial representation quality and temporal modeling choices. For the spatial encoder, incorporating positional information is critical: removing positional encodings leads to a substantial degradation (RMSE 3.64), while fixed sinusoidal encodings significantly outperform learnable variants, suggesting a stronger inductive bias and improved generalization under limited data. The choice of patch size further reveals a trade-off between resolution and abstraction, with a patch size of 16 achieving the best performance, whereas both smaller (overly local) and larger (overly coarse) patches degrade accuracy. Token reduction also plays an important role: moderate reduction (stride 2) yields the lowest error, while both excessive reduction and no reduction harm performance, indicating the need to balance efficiency and spatial fidelity.

Table 7 highlights the importance of temporal modeling. Classical regressors such as Random Forest and XGBoost underperform neural sequence models, indicating that capturing temporal dependencies is essential. While LSTM and attention-based models provide moderate improvements, the proposed GRU-based iterative refinement achieves a substantial gain, reducing RMSE to 1.64. This suggests that modeling temporal evolution via incremental updates is more effective than direct prediction, particularly in low-data regimes. Overall, these ablations confirm that both high-quality spatial representations and carefully designed temporal dynamics are necessary for strong performance.